6.3. Triacylglycerols or Triglycerides

Triacylglycerols are simple lipids synthesized by esterification of glycerol with three molecules fatty acids. It is the storage form of fat, stored in adipose tissue. They are hydrolyzed in the gut by lipases to free fatty acids and monoglycerides.

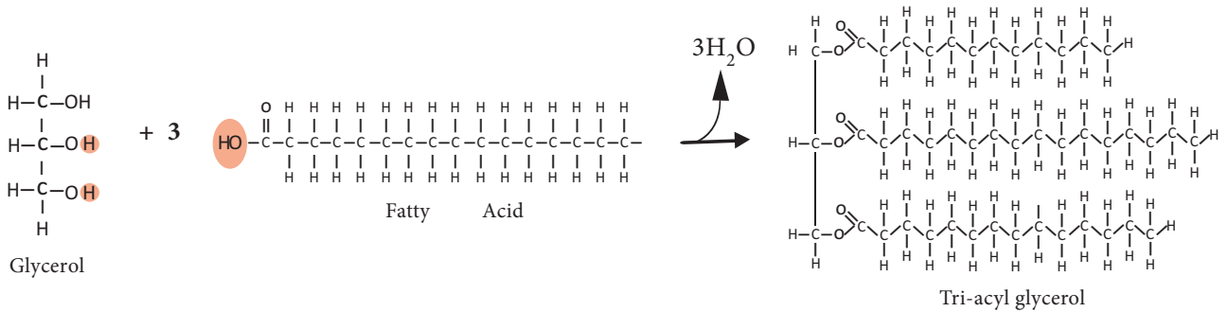


Figure 6.6 Synthesis of triglycerides

If the three hydroxyl groups of glycerol molecule are esterified with the same type of fatty acid then the lipid is called simple triacyl glycerol eg: tripalmitin. If esterification is with different fatty acids, it is called as mixed glycerides eg: dioleopalmitin.

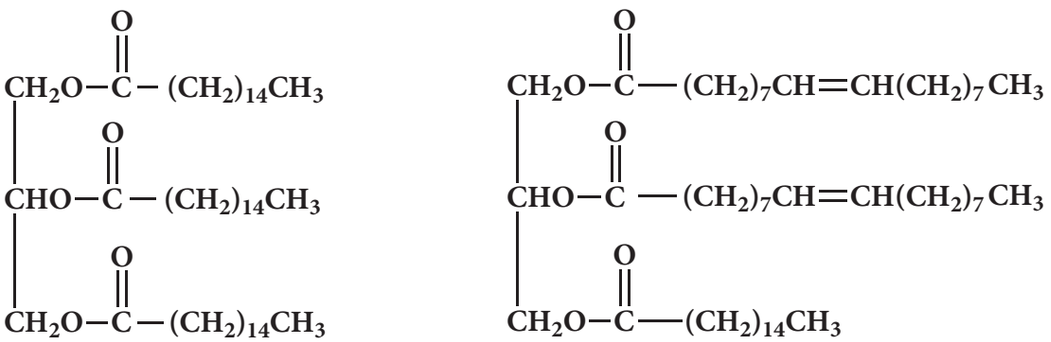


Figure 6.7 Simple and mixed triglycerides

6.3.1. Properties

Physical properties

- They are non-polar, hydrophobic, insoluble in water and soluble in organic solvents.
- Specific gravity is less than water. Therefore fats and oil float on water.

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